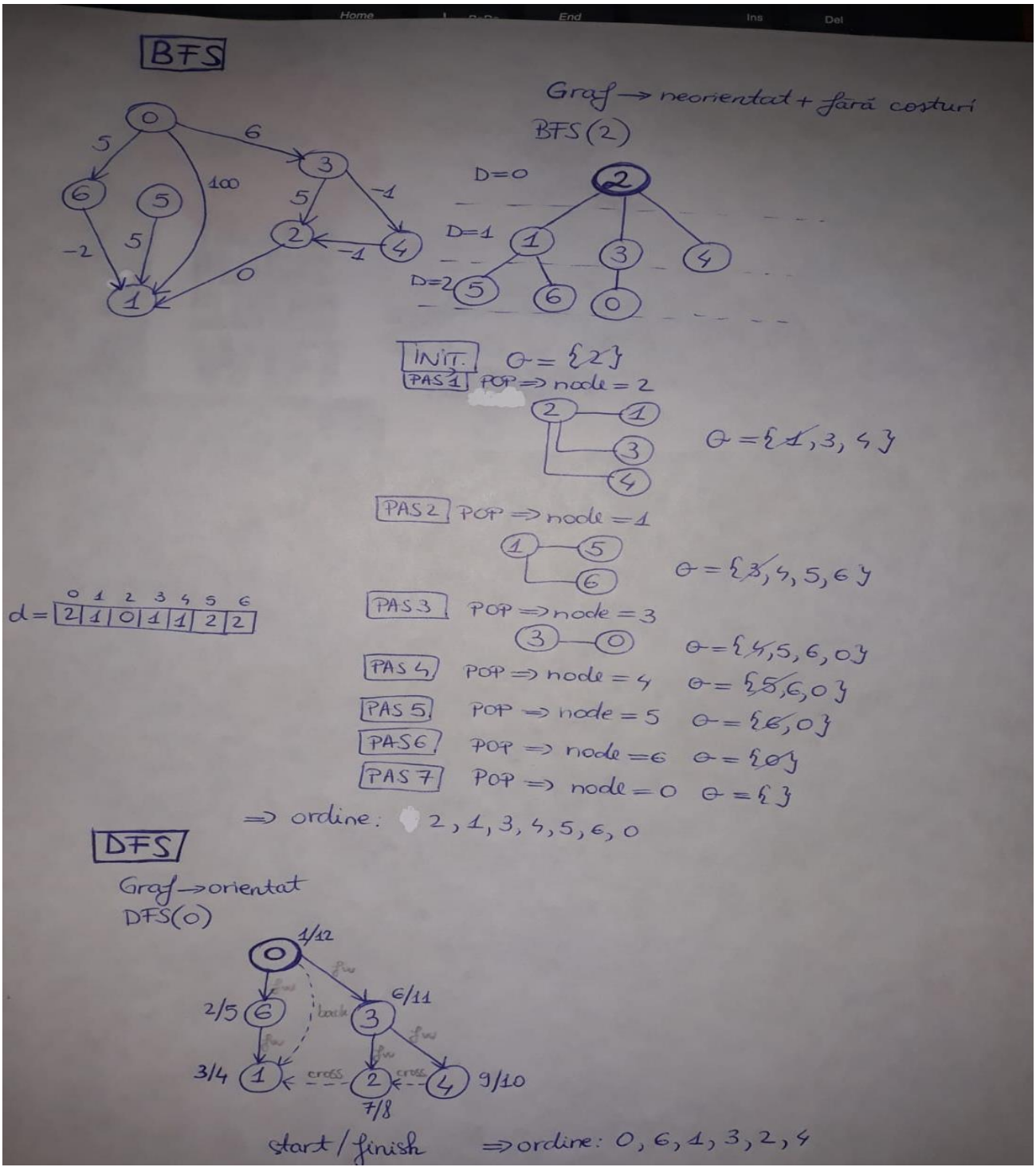
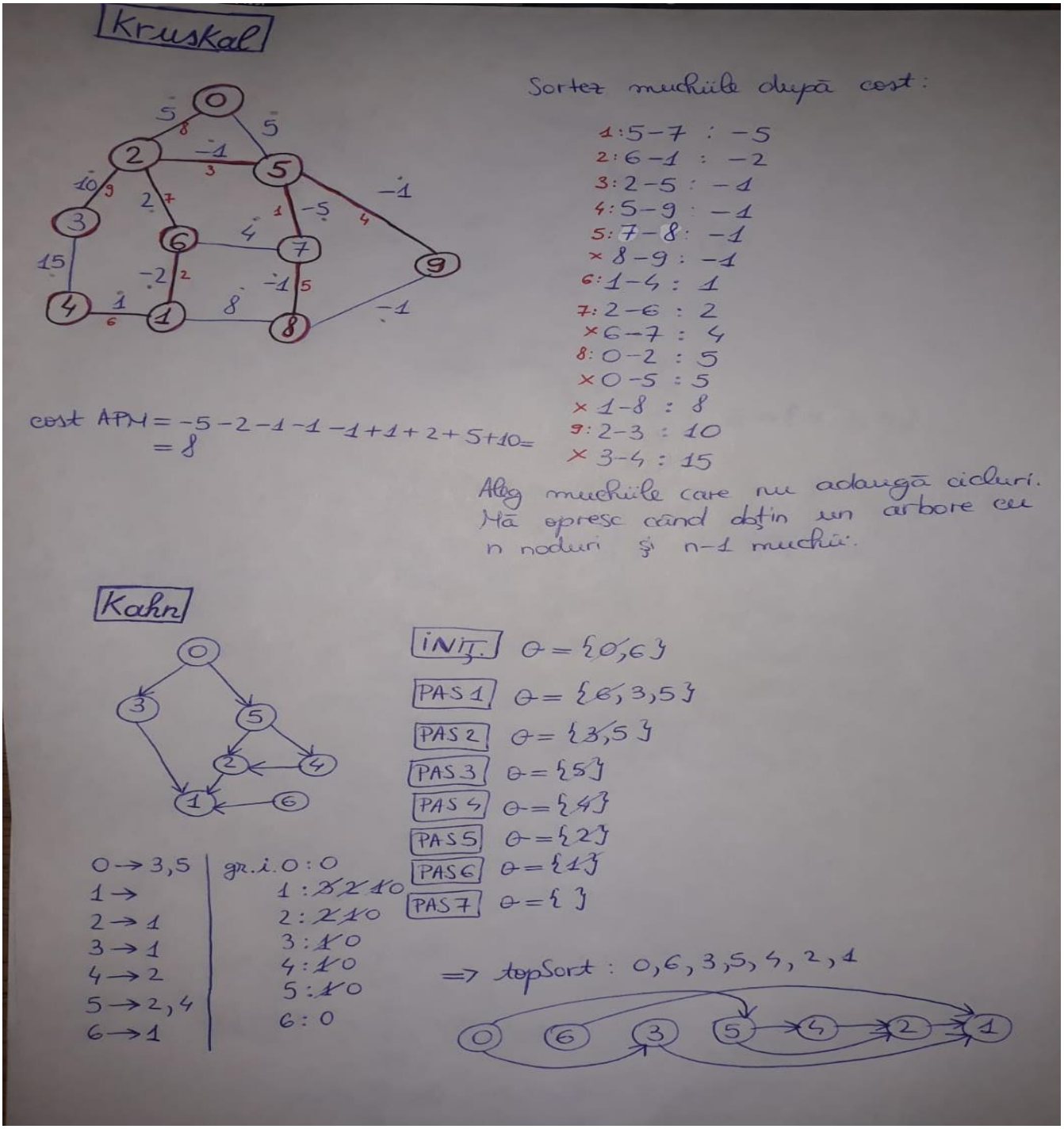
Redactari – subiectul 1

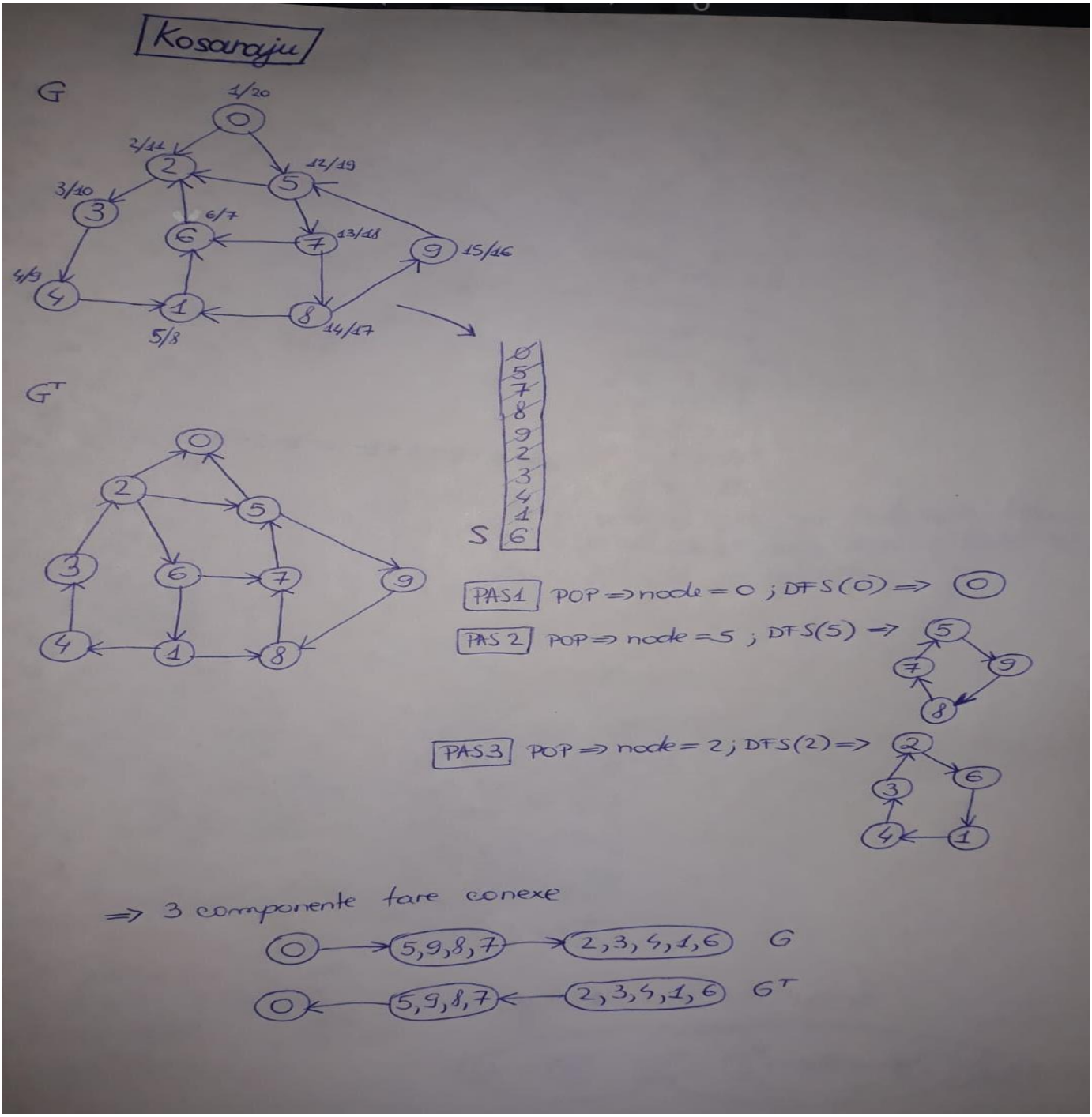
DFS & BFS



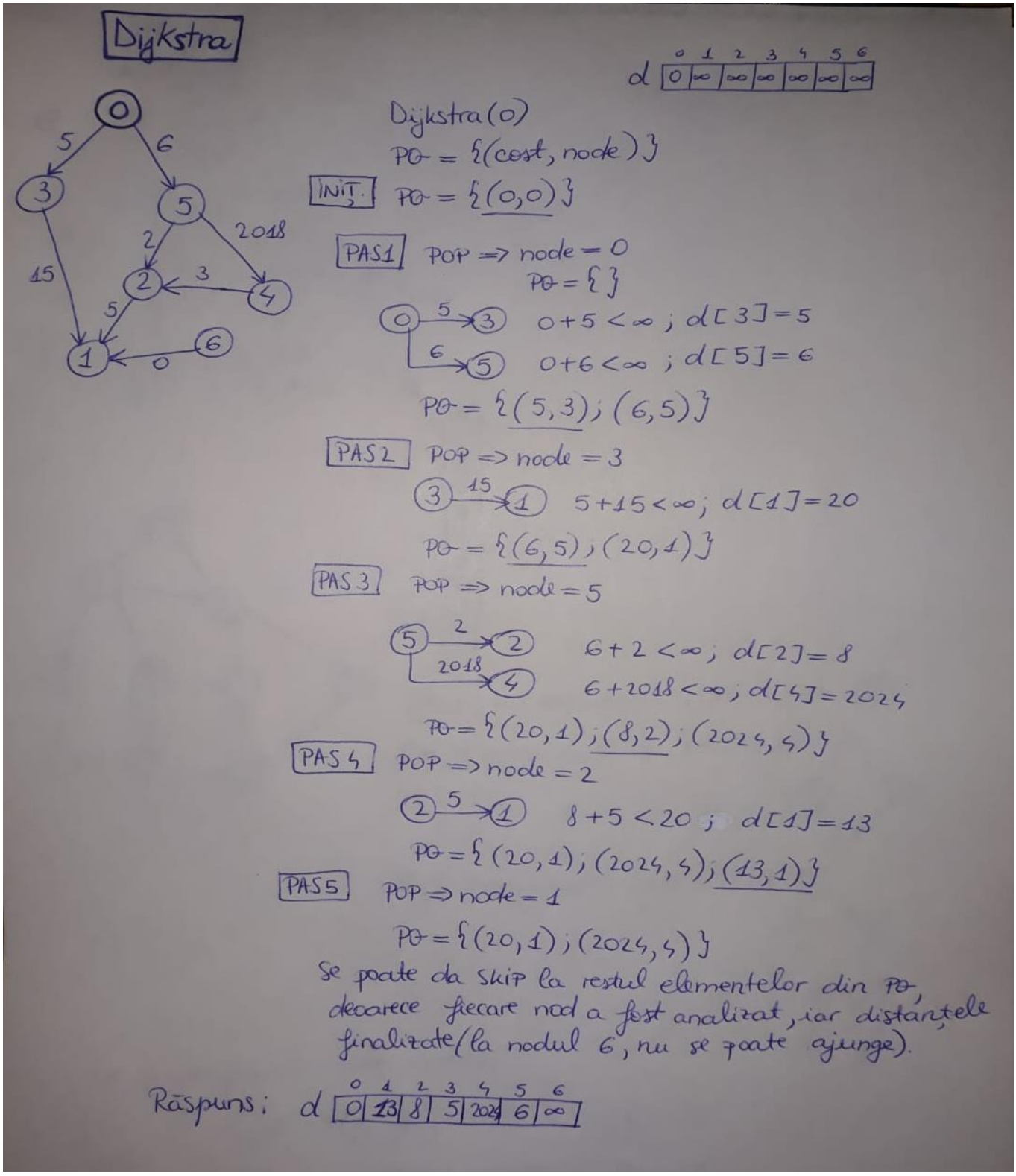
Kruskal (arbore partial de cost minim) & Kahn (sortare topologica)



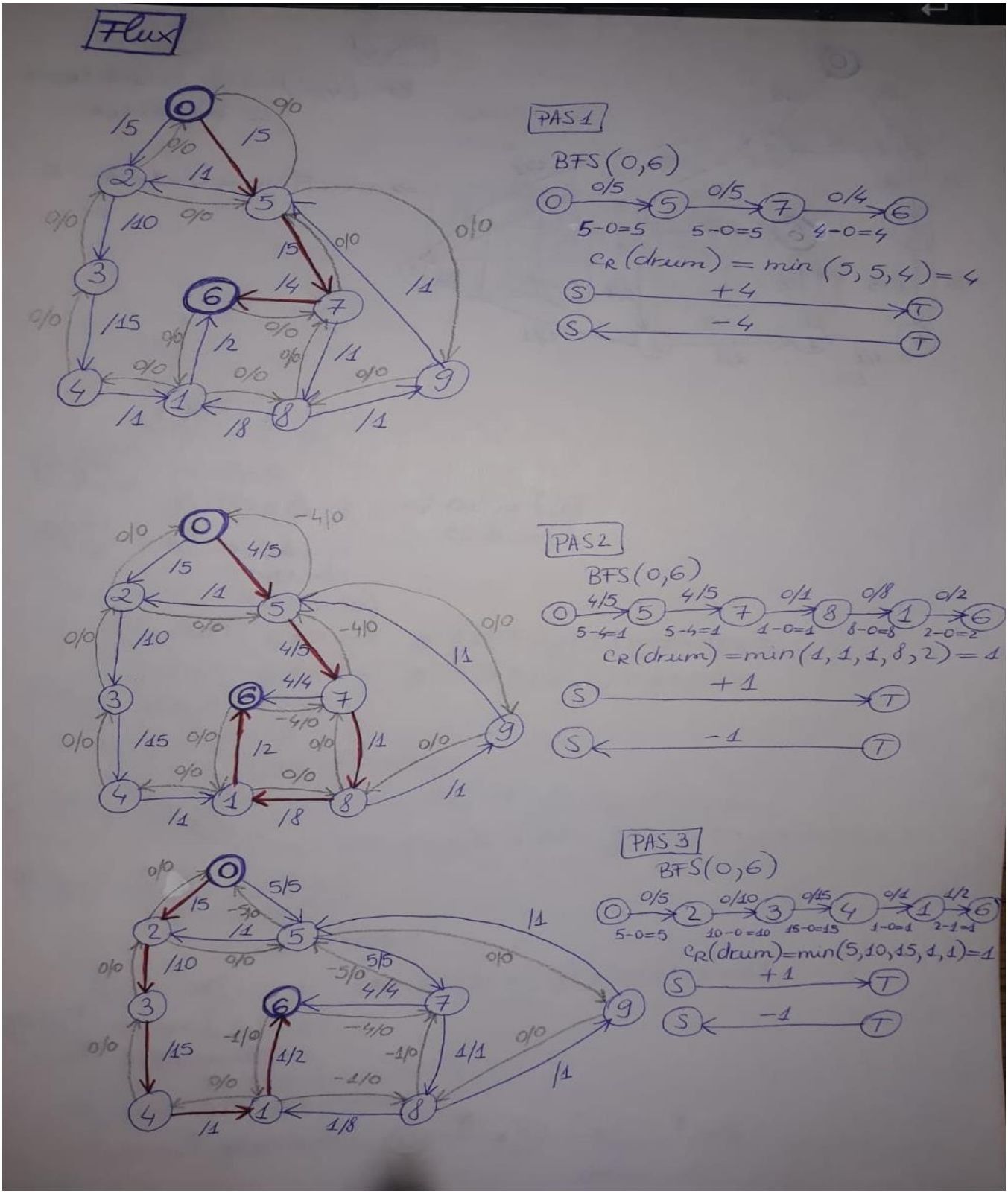
Kosaraju (componente tare conexe)



Dijkstra (distante minime intre sursa si noduri)

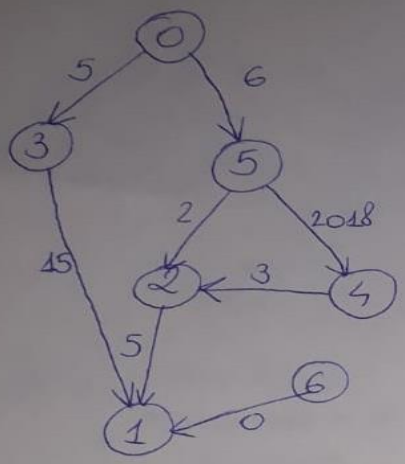


Flux Maxim


Flux
PAS 1
BFS(0,6)
0/5
0/5
0/4
5-0=5
5-0=5
4-0=4
$c_R(drum) = min(5,5,4) = 4$
+4
-4
PAS 2
BFS(0,6)
4/5
4/5
0/1
0/8
0/2
5-4=1
5-4=1
1-0=1
8-0=8
2-0=2
$c_R(drum) = min(1,1,1,8,2) = 1$
+1
-1
PAS 3
BFS(0,6)
0/5
0/10
0/15
0/1
1/2
5-0=5
10-0=10
15-0=15
1-0=1
2-1=1
$c_R(drum) = min(5,10,15,1,1) = 1$
+1
-1

A*

A*



S=0
T=1

	0	1	2	3	4	5	6
g	~~∞~~ 0	~~∞~~ 13	~~∞~~ 8	~~∞~~ 5	~~∞~~ 2024	~~∞~~ 6	∞
h	5	0	5	15	3	2	0
f	5	13	13	20	~~2024~~	8	

INIȚIAL: $g[0]=0$
$f[0]=0+5=5$ $\Rightarrow$ OPEN = {0}
CLOSED = { }

PAS 1: POP() $\Rightarrow$ node = 0
OPEN = { }
CLOSED = {0}

0 —5→ 3: $0+5<\infty$ $g[3]=5$; $f[3]=5+15=20$
0 —6→ 5: $0+6<\infty$ $g[5]=6$; $f[5]=6+2=8$

PAS 2: OPEN = { 3, 5 }
POP $\Rightarrow$ node = 5
CLOSED = {0, 5}

5 —2→ 2: $6+2<\infty$; $g[2]=8$; $f[2]=8+5=13$
5 —2018→ 4: $6+2018<\infty$; $g[4]=2024$; $f[4]=2024+3=2027$

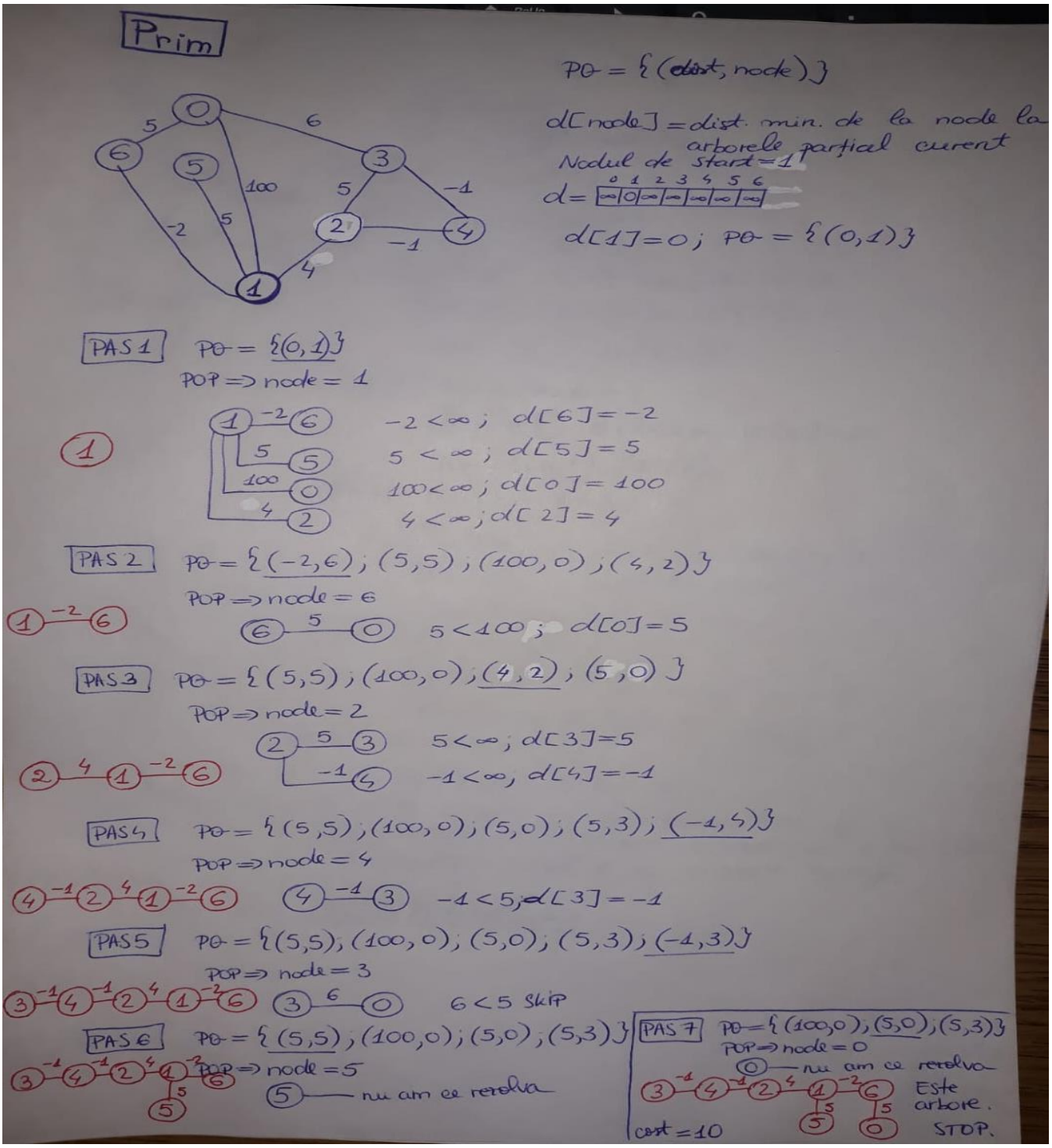
PAS 3: OPEN = { 3, 2, 4 }
POP $\Rightarrow$ node = 2
CLOSED = {0, 5, 2}

2 —5→ 1: $8+5<\infty$; $g[1]=13$; $f[1]=13+0=13$

PAS 4: OPEN = { 3, 4, 1 }
POP $\Rightarrow$ node = 1
CLOSED = {0, 5, 2, 1}
1 = destinație $\Rightarrow$ STOP

Drumul găsit este: 0 —6→ 5 —2→ 2 —5→ 1
costul = 13

Prim (arbore partial de cost minim)



Roy Floyd (distanta minima intre noduri)

